

Relax! This test is not nearly as important as the start of Game of Thrones in nine days!

Each question is worth 2 points except where noted. Good luck!

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

1. A fungus is a type of
 - A. chemoheterotroph**
 - B. photoheterotroph
 - C. photoautotroph
 - D. chemoautotroph
 - E. autoheterotroph
 2. The transfer of fluid from the glomerulus to Bowman's capsule
 - A. is mainly a consequence of blood pressure in the capillaries of the glomerulus.**
 - B. is very selective as to which protein-sized molecules are transferred.
 - C. usually includes the transfer of red blood cells into Bowman's capsule.
 - D. transfers large molecules as easily as small ones.
 - E. results from active transport.
 3. Which of the following statements is correct about diffusion in or out of a cell?
 - A. It requires an expenditure of energy by the cell.
 - B. It is an active process in which molecules move from a region of lower concentration to one of higher concentration.
 - C. It requires aquaporins in the cell wall.
 - D. It requires channel proteins in the cell membrane.
 - E. It is a passive process in which molecules move from a region of higher concentration to a region of lower concentration.**
 4. Birds secrete uric acid as their nitrogenous waste because uric acid
 - A. is readily soluble in water.
 - B. excretion allows birds to live in desert environments.
 - C. requires little water for nitrogenous waste disposal, thus reducing body mass.**
 - D. is metabolically less expensive to synthesize than other excretory products.
 - E. is less toxic than urea.**
 5. A species of rabbit that follows Bergmann's rule
 - A. would be larger in Canada than in Texas.**
 - B. would be smaller in Minnesota than in Florida.
 - C. would have larger ears in Maine and smaller ears in Georgia.
 - D. would be whiter in winter and browner in summer.
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6. Which of the following is *not* a major activity of the stomach?
 - A. Mucus secretion.
 - B. Enzyme secretion.
 - C. Mechanical digestion.
 - D. Nutrient absorption.**
 - E. HCl secretion.
7. Which of the following is an accurate statement about thermoregulation?
 - A. Endotherms are warm-blooded and ectotherms are cold-blooded.
 - B. Endothermy has a lower energy cost than ectothermy.
 - C. Endotherms and ectotherms differ in their primary source of heat for thermoregulation.**
 - D. Endotherms are regulators and ectotherms are conformers.
 - E. Endotherms maintain a constant body temperature and ectotherms do not.
8. Because the foods eaten by animals are often composed largely of macromolecules, this requires the animals to have mechanisms for _____ not found in other groups.
 - A. chemical digestion.**
 - B. filtration in Bowman's capsule.
 - C. absorption of nutrients.
 - D. reabsorption of solutes from the loop of Henle.
 - E. elimination of nitrogenous waste.
9. Which of the following contains digestive enzymes?
 - A. lysosome**
 - B. food vacuole.
 - C. guard cells.
 - D. contractile vacuole.
 - E. central vacuole.
10. Stomach cells are moderately well adapted to the acidity and digesting activities in the stomach by having
 - A. a sufficient colony of *H. pylori* to breakdown the cellulose from vegetables.
 - B. a thick, mucous secretion and active replacement of epithelial cells.**
 - C. a high level of bile.
 - D. secretions enter the stomach from the pancreas.
 - E. a high level of secretion of pepsin.

11. When the digestion and absorption of carbohydrates results in more energy-rich molecules than are immediately required by an animal, the excess is
- A. stored as starch.
 - B. eliminated in the feces.
 - C. stored as glycogen or fat.**
 - D. stored as cellulose or chitin.
 - E. stored in the crop.
12. Upon activation by stomach acidity, the secretions of the gastric glands
- A. delay digestion until the food arrives in the small intestine.
 - B. include nitrogenase and amylase.
 - C. initiate the chemical digestion of lipids in the stomach.
 - D. initiate the mechanical digestion of lipids in the stomach.
 - E. initiate the digestion of protein in the stomach.**
13. After ingestion by humans, the first type of macromolecules to be chemically digested by enzymes in the mouth is
- A. minerals.
 - B. carbohydrates.**
 - C. cholesterol and other lipids.
 - D. proteins.
 - E. nucleic acids.
14. Which of the following membrane activities requires energy from ATP?
- A. movement of water into a cell.
 - B. movement of glucose molecules into a bacterial cell from a solution containing a higher concentration of glucose than inside the cell.
 - C. movement of Na^+ ions from a lower concentration in a mammalian cell to a higher concentration in the extracellular fluid.**
 - D. diffusion of chloride ions across the membrane through a chloride channel.
 - E. movement of water out of a paramecium.
15. Ingested dietary substances must cross cell membranes to be used by the body, a process known as
- A. elimination.
 - B. hydrolysis.
 - C. digestion.
 - D. ingestion.
 - E. absorption.**
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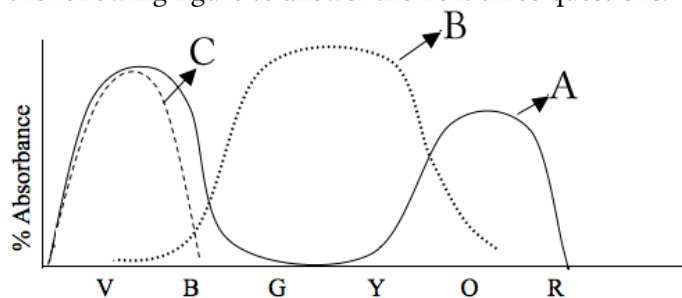
16. Which of the following organs is incorrectly paired with its function?
 - A. Stomach-protein digestion.
 - B. Pancreas-enzyme production.
 - C. Large intestine-bile production.**
 - D. Mouth-starch digestion.
 - E. Small intestine-nutrient absorption.
17. Organisms that obtain energy by oxidizing iron compounds and obtain carbon from CO₂ are called
 - A. autoheterotrophs.
 - B. chemoautotrophs.**
 - C. photochemotrophs.
 - D. chemoheterotrophs.
 - E. photoautotrophs.
 - F. photoheterotrophs.
18. A marine iguana is basking on a sunny rock after feeding in the cold Pacific Ocean water. It's warming its body through
 - A. convection and evaporation.
 - B. conduction and evaporation.
 - C. conduction and radiation.**
 - D. convection and radiation.
 - E. conduction and convection
19. Cell walls are found in at least some representatives of
 - A. plants
 - B. protists
 - C. all of the above.**
 - D. bacteria
 - E. algae
 - F. fungi
20. Small mammals like mice use more energy per unit mass than large mammals mostly because most of the energy is used for
 - A. growing quickly.
 - B. reproduction.
 - C. the basal metabolic rate.
 - D. activities like hunting.
 - E. thermoregulation.**

21. Pepsin is a digestive enzyme that
- A. digests polysaccharides to monosaccharides in the small intestine.
 - B. present in bile to digest lipids in the small intestine.
 - C. begins the digestion of proteins in the stomach.**
 - D. secreted by the pancreas to digest proteins in the duodenum.
 - E. digests nucleic acids in the stomach.
22. Which of the following types of molecules are the major structural components of the cell membrane?
- A. proteins and collagen.
 - B. phospholipids and cellulose.
 - C. proteins and cellulose.
 - D. nucleic acids and proteins.
 - E. phospholipids and proteins.**
23. The advantage of excreting nitrogenous wastes as urea rather than as ammonia is that
- A. less nitrogen is removed from the body.
 - B. urea is less toxic than ammonia.**
 - C. urea can be exchanged for Na^+ .
 - D. urea does not affect the osmoregulation.
 - E. urea requires more water for excretion than ammonia.
24. Stomata will open when
- A. the guard cells have a higher solute concentration than outside the cell.**
 - B. the guard cells have a lower solute concentration than outside the cell.
 - C. the pressure potential inside the guard cell is negative.
 - D. the water potential inside the guard cell is higher than outside the cell.
 - E. the solute potential inside the guard cell is positive.
25. The osmoregulatory/excretory system of a freshwater flatworm is based on the operation of
- A. metanephridia.
 - B. Malpighian tubules.
 - C. nephrons.
 - D. protonephridia.**
 - E. ananephredia.
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Circle True or False. Read each statement completely before answering.

26. True **FALSE** The sources of metabolic nitrogen in the diet are proteins and carbohydrates.
27. True **FALSE** Micronutrients are smaller molecules than macronutrients.
28. True **FALSE** The central vacuole is used by protists to eliminate excess water from the cell.
29. **TRUE** False The primary function of the large intestine is the absorption of water.
30. **TRUE** False A flaccid plant cell is most likely to occur in an isotonic environment.
31. True **FALSE** Cation exchange occurs when plants release CO₂ into the soil to increase the acidity.
32. True **FALSE** A euryhaline species cannot survive in a hypotonic environment.
33. **TRUE** False The chemical energy stored in carbohydrates is a form of potential energy.
34. True **FALSE** Of the six organismal groups we are comparing, only the bacteria and the protists are not monophyletic.
35. **TRUE** False All organisms are considered open systems.

Use the following figure to answer the next three questions.

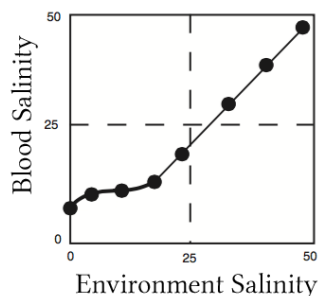


36. Line A represents the absorption spectrum of which group of photopigments? **carotenoids**
37. Line B represents the absorption spectrum of which group of photopigments? **chlorophyll**
38. Line C represents the absorption spectrum of which group of photopigments? **phycobilins**
39. (4 points) Write a *balanced* chemical equation for photosynthesis. $6 \text{CO}_2 + 6 \text{H}_2\text{O} \longrightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6 \text{O}_2 + 6 \text{H}_2\text{O}$
Subtract 1 pt if one prefix number is missing or wrong (e.g., 4 H₂O, 2 pts if more than 1 is wrong.
Subtract 1 pt if one chemical formula is wrong (e.g., C₁₂H₆O₆), 2 pts if one missing more more than 1 wrong.
Ask me if uncertain.

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

Aquaporin	Hypertonic	Peristalsis
Amylase	Hypotonic	Potential
Bile	Iron bacteria	Sodium potassium pump
Chitin	Isotonic	Sphincter
Cellulose	Kinetic	Stomata
Enzymes	Methanogens	Sulfur bacteria
Flame bulb	Nephridopore	Vitamins
Glycogen	Nephridostome	Waxy cuticle
Guard cells	Pancreas	
Hydrochannel	Pepsin	

40. **Enzymes** Cofactors enhance the reaction ability of these macromolecules.
41. **Waxy Cuticle** Prevents water evaporation from most of the leaf surface.
42. **Hypertonic** The tonicity of the environment when Ψ of the environment is lower than Ψ in the organism.
43. **ANY ANSWER** Found in the digestive tract of humans.
44. **Chitin** This structural molecule is found in cell walls and crab shells.
45. **Amylase** The first digestive enzyme secreted by most animals during the digestive process.
46. **Aquaporin** Protein channel that increases water movement into cells.
47. **Kinetic** The energy of motion.
48. **Nephridostome** Opening through which body fluids enter the metanephridia.
49. The following figure shows the blood salinity level of a marine worm compared to the salinity of the surrounding environment. Based on this figure this organism is most likely an



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